

Outage Overview and Restoration State

OUTAGE	FEEDER	START	NORMAL	CUSTOMERS
OI-26-731	12R	2026-07-02 15:42	19:06	8,960

Final cause

Final investigation identifies failed polymer cutout F-12B on Maple lateral. Early draft tree-contact note is superseded but remains visible context.

Metric	Value	Note
Hospital restored	17:18	via T-7 backfeed
All normal	19:06	R12 closed
DER-44	curtailed 17:11	before T-7 close
Cause	F-12B failed cutout	not tree contact
Cutout tag	F12B-882	sent to asset lab
Critical notices	hospital, traffic, LS-4	tracked in notice log

SCADA Alarm Sequence

Time	Device	State	Note
15:42:11	R12 breaker	trip	phase B ground
15:42:14	Recloser 12R-3	lockout	third shot
15:42:19	Relay 50G	pickup	7.8 kA momentary
15:43:02	Cap bank CB-4	offline	voltage sag
15:44:33	OMS	8,960 out	nested outage created
16:05:20	Switch S-18	opened	crew isolation
16:31:05	Switch S-22	opened	Riverside isolated
17:11:00	DER-44	0 kW	curtailment verified
17:18:44	Tie T-7	closed	backfeed hospital
18:05:02	F-12B	replaced	Maple lateral test passed
19:06:10	R12 breaker	closed	normal restored

Control note

Alarm order matters; R12 trip precedes 12R-3 lockout.

Switching Log

Step	Time	Action	Crew	Customers restored
1	16:05	open S-18	Crew A	0
2	16:31	open S-22	Crew B	0
3	16:47	test Maple lateral	Crew A	0
4	17:11	verify DER-44 at 0 kW	Control	0
5	17:18	close T-7	Crew C	4,820
6	18:05	replace cutout F-12B	Crew A	6,140
7	18:42	patrol Riverside taps	Crew B	6,140
8	19:06	close R12 breaker	Control	8,960
9	19:32	normalize T-7	Crew C	all stable

Control note

Step 3 closes T-7 and restores hospital before full feeder restoration.

Customer Impact by Segment

Segment	Customers	Critical load / note	Restored
Maple lateral	2,820	0	18:05
Maple care home spur	86	assisted living	18:05
Hospital loop	1,240	1 hospital	17:18
Riverside apartments	3,100	2 elevators	19:06
Industrial park	1,800	0	19:06
Traffic signal cabinet TS-12	1 service	intersection	17:18
Lift station LS-4	1 service	wastewater	19:06

Recipient	Segment	Time	Method	Message state
Hospital facility desk	Hospital loop	16:04	phone	ETA pending
Hospital facility desk	Hospital loop	17:21	phone	restored via T-7
City traffic	TS-12	17:25	radio	signal power back
Wastewater duty officer	LS-4	18:18	phone	awaiting feeder close
Industrial park contact	Industrial park	19:09	email	normal restored
Care home manager	Maple care home spur	18:08	phone	restored after F-12B change

Control note

Hospital loop is critical and restored at 17:18.

Evidence and Cause Table

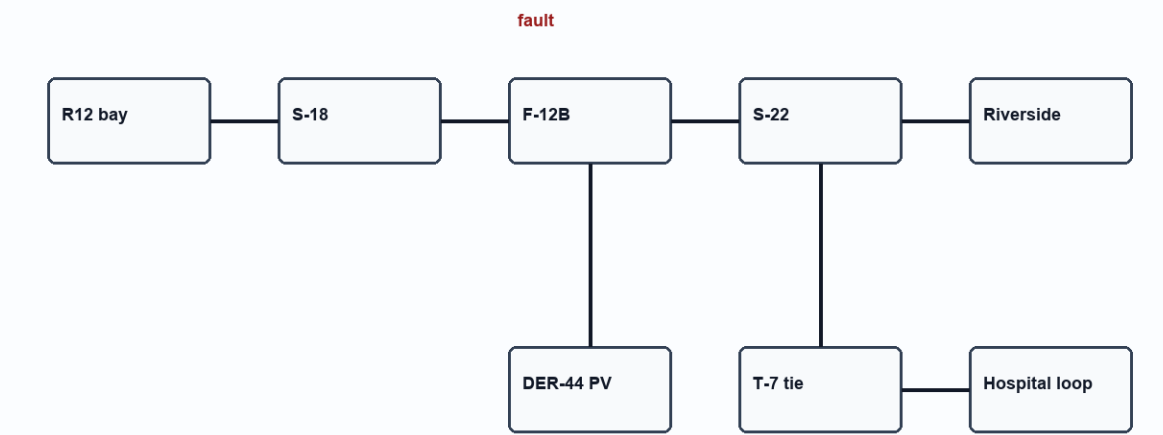
Evidence	Visible value	Finding
Weather radar	cell passed 15:36-15:50	wind gust context
Patrol photo P1	char mark at F-12B	failed cutout
Patrol photo P2	clear span; no branch contact	tree cause superseded
SCADA	phase B ground	matches F-12B
Relay oscillography	single phase-to-ground impulse	not lightning signature
DER log	DER-44 curtailed 17:11	safe backfeed
Cutout tag	F12B-882 polymer barrel split	removed for lab
Draft report	tree contact	superseded
Final report	failed polymer cutout F-12B	controls

Lab item	Serial / tag	Disposition	Reviewer
Cutout barrel	F12B-882	polymer split; archive sample	Protection
Fuse link	Kearney 65K	melt pattern consistent	Asset Eng
Crossarm mark	none	no flashover trace	Patrol
Tree sample	not collected	not applicable	Vegetation

Control note

Final report supersedes the draft tree-contact note.

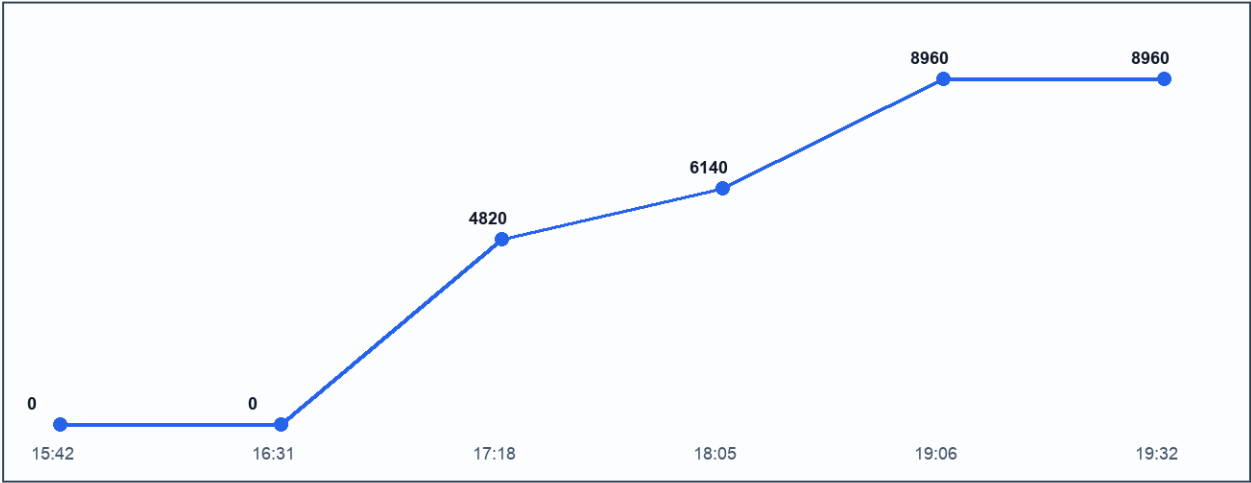
Feeder One-Line Diagram



Node	Relation	Status
Substation Bay R12	feeds 12R trunk	tripped then restored
S-18	upstream sectionalizer	opened for isolation
F-12B	tap fuse to Maple lateral	failed cutout
T-7	tie to feeder 9Q	closed for hospital backfeed
Hospital loop	critical load	restored 17:18

PV site DER-44	backfeed risk	curtailed before close	
Switch / device	Normal	During outage	Field tag
S-18	closed	opened 16:05	yellow hold card
S-22	closed	opened 16:31	visible open blade
T-7	open	closed 17:18	red backfeed tag
DER-44	online	0 kW curtailed 17:11	SCADA clearance
F-12B	energized	failed / replaced	char mark photo P1

Restoration Chart and Weather Panel



Weather item	Value	Interpretation
Radar cell	15:36-15:50	context only
Peak gust	46 mph	contributed stress
Lightning	none within 5 miles	not cause
Tree ticket	draft only	superseded by F-12B finding

Crew Patrol Photo Sheet

Patrol images are stored in the field-media repository. This regulatory copy carries the photo register, GPS/time metadata, reviewer notes, and disposition used for cause determination.

Photo	File	GPS / time	Reviewer note	Use
P1	MG-12R-731-P1	45.5221,-122.681 / 16:18	F-12B char mark	supports failed cutout
P2	MG-12R-731-P2	45.5224,-122.680 / 16:23	clear span; no tree contact	supersedes draft tree cause
P3	MG-12R-731-P3	S-18 / 16:31	open tag visible	isolation evidence
P4	MG-12R-731-P4	T-7 / 17:18	red backfeed tag visible	hospital restoration evidence

Evidence question	Photo reference	Finding	Final treatment
Was a tree touching the line?	P2	No	draft cause superseded
Where is the failed device?	P1	F-12B cutout	final cause
Was S-18 isolated?	P3	open tag	confirmed
Was T-7 used for hospital?	P4	backfeed tag	confirmed

Media-register note

The field-media files are not printed in this copy, but the photo register and reviewer notes are part of the signed evidence packet.

DER Backfeed Clearance

Check	Value	State
DER-44 output	0 kW at 17:11	curtailed
Visible open point	S-22 open	confirmed
T-7 close	17:18	allowed after clearance
Reverse power alarm	none	safe

Clearance item	Operator entry	Field confirmation	Release state
DER-44 disconnect	open 17:06	visible at POI	accepted
SCADA output	0 kW at 17:11	telemetry stable 5 min	accepted
Feeder 9Q load	74% after transfer	under emergency rating	accepted
Reverse power	none	relay 67P clear	accepted
Hospital ATS	normal source lost	T-7 source accepted	accepted
T-7 permission	issued 17:16	close allowed 17:18	accepted
DER restore	after normalize	19:34 call to site	deferred
Site acknowledgment	R. Valdez 19:36	DER owner confirms	logged

Clearance-control note

The T-7 close is valid only after both SCADA output and field-visible disconnect are accepted.
DER restore is deferred until after feeder normalization.

Draft Report Excerpt

Draft field	Draft value	Final value	
Cause	tree contact	failed polymer cutout F-12B	
Hospital restore	18:05	17:18	
DER status	not mentioned	curtailed 17:11	

Revision item	Draft entry	Reviewer mark	Final treatment
Cause	tree contact	crossed in red	failed polymer cutout F-12B
Hospital restore	18:05	circled time mismatch	17:18 via T-7
DER status	not mentioned	blue margin note	curtailed 17:11
Weather	primary cause	downgraded	context only
Photo P2	not referenced	added by reviewer	no tree contact

Revision-control note

This draft excerpt remains in the packet to show the reviewer corrections. The final investigation signoff controls cause, hospital restoration time, and DER status.

Final Investigation Signoff

Signer	Role	Visible state
N. Ortega	Distribution Ops	signed 2026-07-03 09:12
R. Mehta	Protection Eng	signed 2026-07-03 10:44
S. Kim	Regulatory	signed 2026-07-03 11:20
Tree crew	not required	blank

Review gate	Owner	Visible state	Exception
Ops sequence	N. Ortega	signed 09:12	none
Protection review	R. Mehta	signed 10:44	F-12B curve attached
Regulatory review	S. Kim	signed 11:20	major-event no
Vegetation review	Tree crew	blank	not required
DER review	A. Patel	initialed 10:58	restore after normalize

Signature interpretation

The blank tree-crew line is intentional because vegetation was ruled out. DER review is initialed separately because backfeed clearance controlled the T-7 operation.

Regulatory Notice

Field	Value	Note
Major event	No	below threshold
Critical customer	1 hospital	restored 17:18
Notice due	2026-07-05 12:00	internal
Cause category	equipment failure	polymer cutout

Recipient	Channel	Due / sent	Payload note
Hospital liaison	phone	sent 17:21	backfeed restored
City OEM	email	sent 18:12	critical customer update
State PUC log	portal	due 2026-07-05 12:00	below major threshold
Internal claims	ticket	sent 2026-07-03 08:40	equipment failure category
DER owner	phone	sent 19:34	normalization complete
Asset engineering	work order	sent 2026-07-03 09:05	F12B-882 lab teardown
Vegetation contractor	no dispatch	not required	P2 ruled out tree contact

Notice-routing note

The PUC portal item is due even though the incident is below the major-event threshold.
Hospital liaison notice was already sent after T-7 restoration.

Final Restoration Recap

Control item	Final state	Evidence
Cause	failed F-12B cutout	patrol photo + SCADA
Hospital	restored 17:18	T-7 backfeed
All customers	restored 19:06	R12 breaker close
Draft tree contact	superseded	no tree contact photo
DER-44	curtailed 17:11	safe backfeed

Metric	Final value	Source	Reviewer note
CAIDI window	3h 24m	OMS export	exclude 19:32 normalize
Critical restoration	95 minutes	T-7 close	hospital priority
Customers interrupted	8,960	OMS	same as overview
Cause code	equipment failure	final report	not weather
Follow-up WO	WO-7714	asset replacement	cutout audit route

Recap-control note

The 19:32 normalization step is operating cleanup, not customer restoration. Customer restoration stops at the 19:06 R12 breaker close.